



Finding a percentage with a multiplier

Task A: Reviewing percentages of amounts

1) Work out:

a) 51% of $980 = 499.8$

d) 13% of $140 = 18.2$

b) 23% of $640 = 147.2$

e) 3% of $40 = 1.2$

c) 62% of $120 = 74.4$

f) 9% of $24 = 2.16$

Task B: Using a decimal multiplier

1) Write the calculation showing the decimal multiplier.
You do not need to calculate the answer.

a) 56% of $389 = 0.56 \times 389$

d) 3% of $186.2 = 0.03 \times 186.2$

b) 24% of $484 = 0.24 \times 484$

e) 33.3% of $475 = 0.333 \times 475$

c) 16% of $563.1 = 0.16 \times 563.1$

f) 3.3% of $986 = 0.033 \times 986$

2) Work out:

a) $27\% \text{ of } 560 = 0.27 \times 560 = 151.2$

b) $18\% \text{ of } 244 = 0.18 \times 244 = 43.92$

c) $6\% \text{ of } 94 = 0.06 \times 94 = 5.64$

3) Can you explain the strategy behind each student's working out for finding 48% of 64?

$$4 \times 6.4 = 25.6$$

Alex has found 10% and is multiplying by 4 to give 40%.

$$8 \times 0.64 = 5.12$$

Then Alex has found 1% and is multiplying by 8 to give 8%.

$$5.12 + 25.6 = 30.72$$

He has then summed 40% and 8% to make 48% which is 30.72

Alex

Sam has found 1% and is multiplying by 2 to give 2%.

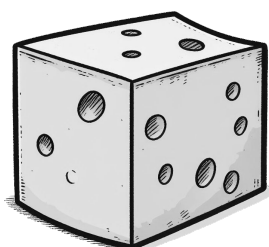
$$2 \times 0.64 = 1.28$$

The 32 represents 50% of the 64.

$$32 - 1.28 = 30.72$$

Sam has subtracted 2% from 50% to make 48% which is 30.72

Sam



4) A hard cheddar cheese contains 37% water, 33% fat, 23% protein, 3.5% carbohydrates, and 3.5% minerals and other compounds.

From a 250g block of cheese, work out how many grams there are of:

a) Water $= 0.37 \times 250 = 92.5 \text{ g}$

b) Fat $= 0.33 \times 250 = 82.5 \text{ g}$

c) Protein $= 0.23 \times 250 = 57.5 \text{ g}$

